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Last time!

Thm (Lagrange) Let p be a prime and

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in \mathbb{Z}[x]$$

with $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has

at most n roots.

Quadratic congruence equations.

We reduced

$$ax^2 + bx + c \equiv 0 \pmod{m}$$

to $y^2 \equiv \Delta \pmod{4am}$, where $\Delta = b^2 - 4ac$ and

$$y = 2ax + b.$$

Quadratic residues & nonresidues modulo $p > 2$.

Defn (Quadratic residues & nonresidues). An integer a , with

$p \nmid a$, is a quadratic residue mod p if there is

$x \in \mathbb{Z}$ s.t. $x^2 \equiv a \pmod{p}$. Otherwise, a is called a

quadratic nonresidue mod p .

Notation: Write $a \in R_p$ if a is a residue mod p ,

$a \in N_p$ if a is a nonresidue mod p .

Lem. Let $p > 2$ be prime and $a \in \mathbb{Z}$ s.t. $p \nmid a$. Then

$$x^2 \equiv a \pmod{p}$$

has no solution or 2 solutions.

The proof shows that if $x \equiv x_0 \pmod{p}$ is a solution, so

$$is \quad x \equiv -x_0 \equiv p - x_0 \pmod{p}.$$

Lemma. Let $p > 2$. There are $\frac{p-1}{2}$ quadratic residues and

$\frac{p-1}{2}$ quadratic nonresidues mod p .

pf. The residues are

$$A: \quad 1^2, 2^2, 3^2, \dots, \left(\frac{p-1}{2}\right)^2,$$

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$$B: \quad (p-1)^2, (p-2)^2, (p-3)^2, \dots, \left(p - \frac{p-1}{2}\right)^2.$$

Take $i^2 \equiv j^2 \pmod{p}$, $1 \leq i, j \leq \frac{p-1}{2}$.

$$\Rightarrow i \equiv j \pmod{p} \quad \text{or} \quad i \equiv -j \pmod{p}$$

As $1 \leq i, j \leq \frac{p-1}{2}$, $|i \pm j| \leq p-1$.

So $i \neq j \Rightarrow 0$, hence $i = j$.

So there are $\frac{p-1}{2}$ residues and $(p-1) - \frac{p-1}{2} = \frac{p-1}{2}$

nonresidues. □

Thm (Euler's criterion - version 1). Let $p \geq 2$ and $a \in \mathbb{Z}$ s.t.

$p \nmid a$.

(1) If $a \in \mathbb{K}_p$, then $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

(2) If $a \in \mathbb{N}_p$, then $a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$.

Pf. (1) If $a \in \mathbb{K}_p$, there is some $x \in \mathbb{Z}$ s.t.

$$x^2 \equiv a \pmod{p}$$

$$\text{Fermat} \Rightarrow a^{\frac{p-1}{2}} \equiv (x^2)^{\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \pmod{p}.$$

(2) Consider

$$\underbrace{x^{p-1} - 1}_{f(x)} = \underbrace{(x^{\frac{p-1}{2}} - 1)}_{g(x)} \underbrace{(x^{\frac{p-1}{2}} + 1)}_{h(x)}.$$

$\left\{ \begin{array}{l} \text{Fermat} \\ \text{Lagrange} \end{array} \right. \Rightarrow f(x) \equiv 0 \pmod{p}$ has exactly $p-1$ roots:

$$x \equiv 1, 2, \dots, p-1 \pmod{p}.$$

Part (1) $\stackrel{\text{Lem}}{\Rightarrow} g(x) \equiv 0 \pmod{p}$ has at least $\frac{p-1}{2}$ roots

given by all the quadratic residues mod p .

$\deg g(x) = \frac{p-1}{2} \stackrel{\text{Lagrange}}{\Rightarrow} g(x) \equiv 0 \pmod{p}$ has at most $\frac{p-1}{2}$ roots.

So $g(x) \equiv 0 \pmod{p}$ has exactly $\frac{p-1}{2}$ roots.

Now, if $a \not\equiv 0 \pmod{p}$, then $f(a) \equiv 0 \pmod{p}$ but $g(a) \not\equiv 0 \pmod{p}$.

So $h(a) \equiv 0 \pmod{p}$, i.e. $a^{\frac{p-1}{2}} + 1 \equiv 0 \pmod{p}$ $\square$

Defn (Legendre symbol) For any prime $p > 2$ and $a \in \mathbb{Z}$, define

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & , \quad a \in \mathbb{R}_p, \\ -1 & , \quad a \in \mathbb{N}_p, \\ 0 & , \quad p \mid a. \end{cases}$$

Thm (Euler's criterion - version 2). For any prime $p > 2$ and

$a \in \mathbb{Z}$ with $p \nmid a$,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

Cor (Properties of Legendre symbol) Let $p > 2$ be prime.

(1) (Periodicity) $a \equiv b \pmod{p} \Rightarrow \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$

(2) (Complete multiplicativity). Given $a, b \in \mathbb{Z}$,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

In particular, if $ab \equiv 1 \pmod{p}$, then

$$\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$$

Pf. Exercise.

Cor $(-1 \pmod{p})$ Let $p > 2$ be prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}. \quad (*)$$

In other words, $-1 \not\equiv p$ if $p \equiv 1 \pmod{4}$;

$-1 \equiv p$ if $p \equiv 3 \pmod{4}$.

Pf. (*) follows from Euler's criterion, version 2.

Note also that $\frac{p-1}{2}$ is even when $p \equiv 1 \pmod{4}$ and odd when $p \equiv 3 \pmod{4}$. For instance, $\frac{p-1}{2}$ is even if

$$\frac{p-1}{2} \equiv 0 \pmod{2}$$

$$\Leftrightarrow p-1 \equiv 0 \pmod{4}$$

$$\Leftrightarrow p \equiv 1 \pmod{4}.$$



E.g. $-1 \nmid 13$, since $13 = 3 \cdot 4 + 1 \equiv 1 \pmod{4}$.

$-1 \nmid 11$, since $11 = 2 \cdot 4 + 3 \equiv 3 \pmod{4}$.

Cor. Let $p \geq 2$ be prime.

(1) (Wilson's thm) $(p-1)! \equiv -1 \pmod{p}$.

(2) Suppose $p > 2$. Let

$$R = \{\text{quadratic residues mod } p\},$$

$$N = \{\text{quadratic nonresidues mod } p\}.$$

$$\prod_{r \in R} r \equiv (-1)^{\frac{p+1}{2}} \pmod{p}$$

$$\prod_{n \in N} n \equiv (-1)^{\frac{p-1}{2}} \pmod{p}.$$

Moreover, if $p \geq 5$, then

$$\sum_{r \in R} r \equiv \sum_{n \in N} n \equiv 0 \pmod{p}.$$